

SAFE-BY-DESIGN SCREENING & GOVERNANCE

SafePro

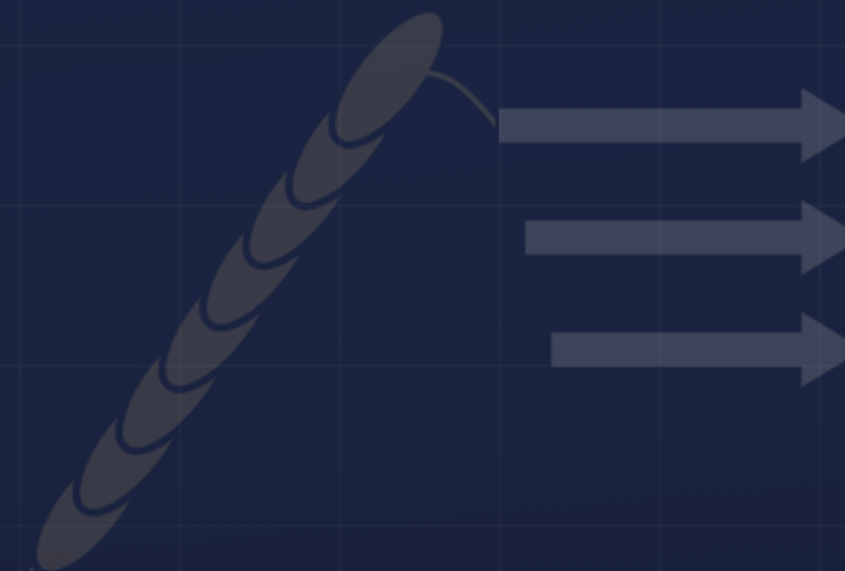
Screening AI-designed proteins for developability and *biosecurity risk*, before synthesis.

TEAM

Dalan

PROBLEM STATEMENT

Computational pre-clinical drug development



01

PROBLEM STATEMENT · COMPUTATIONAL PRE-CLINICAL DRUG DEVELOPMENT

Choosing which candidates to make is the *bottleneck*.

AI designs faster than labs can test. Most fail on whether the molecule can be made.

90%

of clinical candidates fail

ACTA PHARM. SIN. B, 2022

214M+

AI-predicted structures (300k in 2021)

ALPHAFOLD DB, 2024

~\$1B

and 10 to 15 years per drug

ASBMB TODAY, 2022

AI can design toxin-like proteins that slip past *sequence-based* screening.

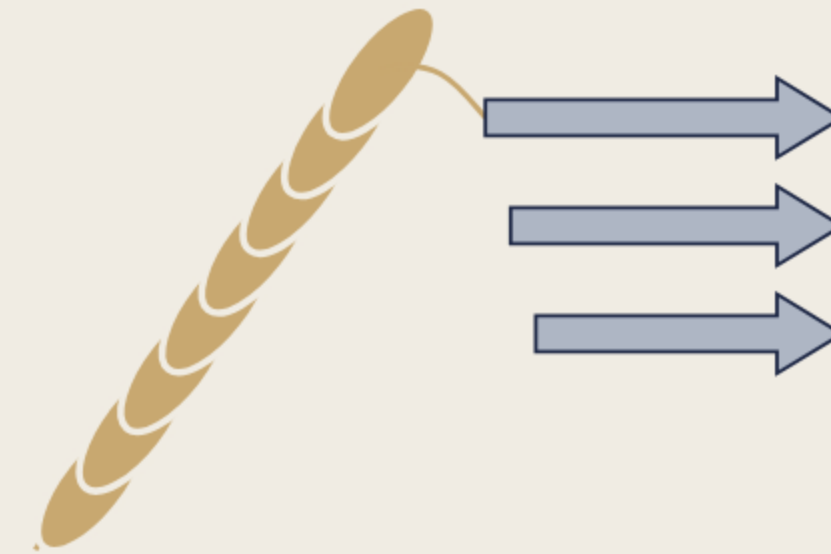
SCIENCE, 2025

WHERE SAFEPRO FITS

A checkpoint *before* synthesis.

SafePro reviews each AI-designed candidate and ranks what's worth making.

- Ranks developability and manufacturability
- Screens biosecurity by structure, not just sequence
- Runs between design and synthesis



AI-DESIGNED THERAPEUTIC PROTEIN

HOW WE BUILD IT

From a sequence to a *defensible* decision.

INPUT

A candidate

Sequence + predicted structure
(*AlphaFold*, ESMFold).



SCORING · TWO MODELS

Developability: aggregation, stability, immunogenicity. (TAP, CamSol, ESM-2)

Biosecurity: structural match to known toxins. (ToxinPred, VFDB)



DECISION · LOGGED

PASS

Cleared

REVIEW

Human review

BLOCK

Held back

PUBLIC TRAINING DATA: PDB · UNIPROT/SWISS-PROT · SABDAB/THERA-SABDAB. EVERY FLAG IS LOGGED.

PHASE 1 · PROTOTYPE

Build both scorers

Both scorers, on public data.

PHASE 2 · VALIDATE

Replay real candidates

Score past candidates, tune thresholds.

PHASE 3 · PILOT

Run it in one team

Internal assay data, one team.

WHY PFIZER

Built for Pfizer's *pipeline* and its values.



PIPELINE

Strengthens Stage 1 discovery

Triage across biologics, vaccines, oncology.



MANUFACTURING

De-risks the supply chain early

Weak designs fail before the Supply Sites.



VALUES

Responsible AI, by design

Audit logs and human review, built in.

DEVELOPABILITY DRIVES MOST BIOLOGIC ATTRITION. COMMS. BIOLOGY, 2024

WHAT STANDS IN THE WAY

The two *hard* parts.

CHALLENGE

Getting good training data



HOW WE ANSWER IT

Public data first, internal data in the pilot

CHALLENGE

False positives



HOW WE ANSWER IT

A human reviews every flag

WHAT CHANGES IF THIS WORKS

Fewer dead ends, and a *safer* shortlist.

- Fewer late failures
- A safer shortlist into the lab
- Review built into the workflow